

PROJECT REPORT
on

**VIDEO RECOMMENDATION SYSTEM
USING ECG SENSOR**

submitted by

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NAAC Accredited with "A" Grade (CGPA : 3.18)



Department of Electronics and Telecommunication Engineering

This is to certify that the Project Report Stage – II

“Video Recommendation System using ECG Sensor”

Submitted by:

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Students of **Electronics and Telecommunication Engineering** have successfully completed their **Project Stage – II** required for the fulfillment of **SEM VIII** as per the norms prescribed by the **University of Mumbai** during the First half of the year 2024. The project report has been assessed and found to be satisfactory.

Internal Guide

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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The successful completion of this project was made possible by the unwavering guidance and support of numerous individuals, and we feel incredibly fortunate to have had their assistance throughout our journey. This project represents the culmination of both the theoretical knowledge and practical skills we have acquired during our degree, and we only hope to do justice to the efforts of those who have contributed to its success.

We extend our heartfelt gratitude to our project guide, Professor Tanaji Biradar, from the Department of Electronics and Telecommunication, for his invaluable mentorship and support throughout this endeavor. We would also like to express our appreciation to the entire department, headed by Dr. Amit Deshmukh, as well as our colleagues, who have generously offered their assistance and support.

Furthermore, we are thankful to Dr. Hari Vasudevan for providing us with the unique opportunity to collaborate on this project. Finally, we extend our gratitude to all individuals who, directly or indirectly, have contributed to the success of this initiative.

Abstract

In this project, we have successfully implemented a system for recommending videos based on a person's mood, utilizing an ECG sensor for real-time mood detection. The proliferation of digital content platforms has created a need for personalized content recommendations aligned with users' emotional states. Our system addresses this need by integrating physiological signals, specifically ECG data, into the video recommendation process.

The key components of our system include data collection using ECG sensors, mood classification using machine learning algorithms, a video dataset from YouTube, a recommendation engine, and a user interface. We have employed various machine learning techniques to classify mood based on ECG signals, achieving the desired level of accuracy for effective video recommendations.

Throughout the project, we have prioritized ethical considerations, ensuring user data privacy and security. We have implemented measures to safeguard user data and have adhered to ethical guidelines in all aspects of our project.

Our system has demonstrated the feasibility and effectiveness of using physiological signals for personalized video recommendations. By providing users with content that resonates with their emotional state, we aim to enhance the user experience on digital content platforms and improve engagement with the recommended videos.

Introduction

Our project focuses on the development of a sophisticated system for recommending videos based on an individual's mood, leveraging the use of physiological signals captured through an ECG sensor. This system represents a novel approach to enhancing user engagement and satisfaction on digital content platforms by offering personalized video suggestions that align with the user's emotional state. The project encompasses the integration of hardware and software components to enable real-time monitoring of the user's mood and the delivery of relevant video recommendations. By combining advanced signal processing techniques with machine learning algorithms, our system aims to provide users with a unique and tailored video viewing experience that adapts to their emotional needs and preferences.

The exponential growth of digital content platforms has led to an unprecedented amount of video content being available to users. While this abundance offers a wide range of choices, it also poses a challenge in terms of content discovery and selection. Users often struggle to find videos that match their current mood or emotional state, resulting in frustration and disengagement. Personalized video recommendations address this challenge by leveraging user data, including viewing history and preferences, to offer content that is more likely to resonate with the user's emotions. This not only enhances the user experience but also increases the likelihood of users discovering new content that they may enjoy, thereby increasing user retention and platform engagement.

The core of our system lies in the use of ECG Monitoring with AD8232 ECG Sensor & ESP32 for real-time monitoring of the user's heart activity. The ECG sensor captures subtle changes in the user's heart rate, which are known to correlate with emotional states such as stress, excitement, and relaxation. The captured ECG data is then processed using the Pan Tomkins Algorithm, a widely used method for ECG analysis, to extract features that characterize the user's mood. These features are then fed into a machine learning model trained to classify the user's mood based on the ECG signals. By combining hardware and software components, our system can accurately detect the user's emotional state and recommend videos that are tailored to their mood, thereby enhancing the overall user experience on digital content platforms.

Literature Survey

The paper by J. Song and S. Kim presents a comprehensive review of existing research on mood detection using ECG signals for personalized music recommendation. The authors highlight the growing interest in using physiological signals for emotion recognition and its potential applications in enhancing user experiences in digital content platforms. They discuss various approaches and algorithms proposed in previous studies for mood detection from ECG signals, including signal processing techniques and machine learning models. The paper also discusses the challenges and limitations of current methods, such as the need for robust feature extraction methods and the importance of addressing individual differences in physiological responses. Overall, the paper provides a valuable overview of the state-of-the-art in mood detection from ECG signals and sets the stage for the development of more effective and personalized recommendation systems in the future. [1]

S. Gupta and R. Sharma presents an in-depth review of the research landscape surrounding ECG signal-based emotion recognition systems for music recommendation. The authors delve into the significance of emotion recognition in enhancing user experiences, particularly in the context of music selection. They explore a variety of methodologies employed in previous studies, including signal processing techniques and machine learning algorithms, for extracting emotional cues from ECG signals. Additionally, the paper discusses the challenges associated with such systems, such as the need for real-time processing and the importance of individual differences in emotional responses. By synthesizing existing literature, the authors provide valuable insights into the current state of research and lay a foundation for future advancements in ECG-based emotion recognition systems for personalized music recommendation. [2]

The paper by A. Roy and P. Das offers a comprehensive review of the literature on mood detection from ECG signals for content recommendation in social media. The authors emphasize the importance of personalized content recommendation in enhancing user engagement and satisfaction on social media platforms. They discuss various approaches and techniques used in previous studies for mood detection, including feature extraction methods and machine learning algorithms. The paper also highlights the challenges associated with mood detection from

physiological signals, such as the need for robust signal processing techniques and the impact of noise and artifacts. Overall, the paper provides a valuable overview of the state-of-the-art in mood detection from ECG signals for content recommendation in social media, laying the groundwork for further research in this area. [3]

M. Li and H. Zhang presents a thorough literature review on emotion recognition from ECG signals for personalized movie recommendation. The authors highlight the significance of emotion recognition in improving user experience and engagement with movie content. They discuss various methodologies and techniques used in previous studies for emotion recognition from ECG signals, including signal processing algorithms and machine learning models. The paper also addresses the challenges associated with emotion recognition from physiological signals, such as the need for accurate feature extraction and the impact of individual differences in emotional responses. Overall, the paper provides a comprehensive overview of the current state of research in emotion recognition from ECG signals for personalized movie recommendation, laying the foundation for future advancements in this field. [4]

The paper by S. Park and J. Lee provides a detailed literature review on real-time emotion recognition systems using ECG signals for video recommendation. The authors emphasize the importance of real-time processing in enhancing user experiences, particularly in the context of video recommendation. They discuss various methodologies and algorithms used in previous studies for emotion recognition from ECG signals, including signal processing techniques and machine learning models. Additionally, the paper highlights the challenges faced by such systems, such as the need for robust feature extraction methods and the impact of noise in ECG signals. By synthesizing existing literature, the authors provide valuable insights into the current state of research and set the stage for future advancements in real-time emotion recognition systems for video recommendation. [5]

The paper by J. Wang and H. Zhang offers an extensive review of the literature on emotion recognition using physiological signals. The authors delve into the importance of physiological signals, such as ECG, in accurately detecting human emotions. They discuss various signal processing techniques and machine learning algorithms used in previous studies for emotion

recognition, highlighting the strengths and limitations of each approach. The paper also addresses the challenges associated with emotion recognition from physiological signals, such as noise interference and the need for real-time processing. Overall, the paper provides a comprehensive overview of the current state of research in emo

tion recognition using physiological signals, serving as a valuable resource for researchers and practitioners in the field. [6]

R. Wang and Y. Li offers a comprehensive survey of the literature on emotion recognition based on ECG signals. The authors highlight the growing interest in using physiological signals for emotion recognition and its potential applications in various fields, including healthcare and human-computer interaction. They discuss the challenges associated with ECG-based emotion recognition, such as the need for robust feature extraction methods and the impact of noise and artifacts. The paper also reviews the different approaches and algorithms proposed in previous studies for emotion recognition from ECG signals, providing insights into the current state of research in this field. Overall, the paper provides a valuable resource for researchers and practitioners interested in ECG-based emotion recognition, highlighting key trends and challenges in the field. [7]

The paper by X. Zhang and L. Wang offers a detailed literature review on emotion recognition from ECG signals for personalized services. The authors emphasize the importance of personalized services in enhancing user experience and satisfaction in various domains. They discuss a range of approaches and techniques used in previous studies for emotion recognition from ECG signals, including feature extraction methods and machine learning algorithms. The paper also highlights the challenges in emotion recognition from physiological signals, such as the need for real-time processing and the impact of noise and artifacts. Overall, the paper provides a comprehensive overview of the current state of research in emotion recognition from ECG signals for personalized services, paving the way for future advancements in this field. [8]

The paper by Y. Chen and S. Zhang provides an extensive review of literature on emotion recognition from ECG signals using wavelet transform and machine learning techniques. The

authors emphasize the importance of emotion recognition in various applications, including healthcare and human-computer interaction. They discuss the advantages of using wavelet transform for ECG signal analysis, such as its ability to capture both frequency and time-domain information. Additionally, the paper reviews the use of machine learning techniques, such as support vector machines and neural networks, for emotion classification based on ECG signals. The authors also highlight the challenges in emotion recognition from ECG signals, such as noise and artifacts, and discuss potential solutions. Overall, the paper offers a comprehensive overview of the current state of research in emotion recognition from ECG signals using wavelet transform and machine learning techniques, paving the way for future advancements in this field. [9]

The paper by S. Lee and H. Kim presents a detailed literature review on mood recognition systems using ECG signals. The authors emphasize the importance of mood recognition in various applications, including personalized content recommendation and mental health monitoring. They discuss the use of ECG signals for mood recognition, highlighting the advantages of using physiological signals over other modalities. The paper also reviews different approaches and algorithms proposed in previous studies for mood recognition from ECG signals, including feature extraction techniques and machine learning models. Additionally, the authors discuss the challenges and limitations of current methods, such as the need for robust and real-time signal processing algorithms. Overall, the paper provides a comprehensive overview of the state-of-the-art in mood recognition systems using ECG signals, laying the groundwork for future research in this area. [10]

A. Patel and B. Patel offers a comprehensive literature review on ECG signal-based emotion recognition systems for movie recommendation. The authors emphasize the growing interest in using physiological signals for emotion recognition and its potential applications in enhancing user experiences in movie recommendation systems. They discuss various methodologies and algorithms proposed in previous studies for emotion recognition from ECG signals, including signal processing techniques and machine learning models. The paper also highlights the challenges and limitations of current methods, such as the need for robust feature extraction methods and the importance of addressing individual differences in physiological responses. Overall, the paper provides a valuable overview of the state-of-the-art in ECG signal-based

emotion recognition systems for movie recommendation, setting the stage for the development of more effective and personalized recommendation systems in the future. [11]

The paper by L. Wu and Z. Wu offers a detailed review of the literature on emotion recognition from ECG signals using deep learning models. The authors underscore the growing interest in leveraging deep learning techniques for extracting emotional cues from physiological signals, particularly ECG signals. They delve into various deep learning architectures and methodologies employed in previous studies, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), for emotion recognition. The paper also discusses the advantages of deep learning models, including their ability to automatically learn relevant features from raw data, as well as the challenges, such as the need for large annotated datasets. Overall, the paper provides a comprehensive overview of the state-of-the-art in using deep learning models for emotion recognition from ECG signals, highlighting their potential in enhancing the accuracy and efficiency of emotion recognition systems. [12]

The paper by S. Sharma and R. Singh provides an extensive literature review on emotion recognition systems using ECG signals and genetic algorithms. The authors emphasize the importance of emotion recognition in various applications, including human-computer interaction and healthcare. They discuss the evolution of emotion recognition techniques, highlighting the shift towards using physiological signals such as ECG for more accurate and reliable emotion detection. The paper also reviews the role of genetic algorithms in optimizing feature selection and classification models for emotion recognition. Overall, the paper offers a comprehensive overview of the state-of-the-art in emotion recognition systems using ECG signals and genetic algorithms, setting the stage for further advancements in this field. [13]

The paper by J. Liu and Y. Liu presents a thorough review of the literature on emotion recognition using ECG signals and convolutional neural networks (CNNs). The authors emphasize the growing interest in using deep learning techniques for emotion recognition, particularly CNNs, due to their ability to automatically learn relevant features from raw ECG signals. They discuss various CNN architectures and training methodologies used in previous studies for emotion recognition from ECG signals. The paper also highlights the advantages of using CNNs, such as their ability to

capture complex patterns in ECG signals, and discusses the challenges and limitations of current CNN-based approaches. Overall, the paper provides a comprehensive overview of the state-of-the-art in emotion recognition using ECG signals and CNNs, laying the groundwork for future research in this area. [14]

The paper by M. Chen and H. Zhang presents a comprehensive study on emotion recognition using physiological signals, focusing on the advancements and challenges in this field. The authors review a wide range of physiological signals, including ECG, EEG, and skin conductance, and their roles in capturing emotional states. They discuss various signal processing techniques and machine learning algorithms employed in previous studies for emotion recognition, highlighting their effectiveness and limitations. The paper also addresses the challenges associated with physiological signal-based emotion recognition, such as the need for robust feature extraction methods and the impact of individual differences in physiological responses. Overall, the paper offers a thorough overview of the current state of research in this area, providing valuable insights for future developments in emotion recognition using physiological signals. [15]

The paper by S. Kumar and R. Gupta presents a novel approach to emotion recognition using ECG signals and Long Short-Term Memory (LSTM) networks. The authors highlight the significance of accurate emotion recognition in various applications, including personalized content recommendation. They discuss the architecture of LSTM networks and their suitability for processing sequential data such as ECG signals. The paper describes the methodology for preprocessing ECG signals and training LSTM networks for emotion recognition. The authors also present experimental results demonstrating the effectiveness of their approach in accurately recognizing emotions from ECG signals. Overall, the paper contributes to the advancement of emotion recognition techniques using physiological signals, offering a promising avenue for future research in this area. [16]

Background

Evolution of Video Content Consumption:

The consumption of video content has undergone a significant transformation in recent years, largely due to the proliferation of digital platforms and the increasing availability of high-speed internet. With platforms like YouTube, Netflix, and Amazon Prime Video offering a vast array of content, users are faced with the challenge of navigating through a sea of videos to find content that aligns with their interests and preferences. This has led to the need for more personalized recommendation systems that can effectively cater to individual user preferences.

Example:

YouTube, founded in 2005, revolutionized the way people consume video content by offering a platform where anyone can upload and share videos. This democratization of content creation has led to a massive increase in the variety and quantity of videos available to viewers.

Another example of the evolution of video content consumption is the rise of live streaming platforms such as Twitch. Originally focused on video game live streaming, Twitch has expanded to include a wide range of content categories, including music, art, and talk shows. This platform has redefined the concept of "broadcasting," allowing creators to interact with their audience in real-time and creating new opportunities for content creators and viewers alike. The success of Twitch has highlighted the growing demand for live and interactive content, further shaping the landscape of video content consumption.

Challenges in Traditional Video Recommendation Systems:

Traditional video recommendation systems often rely on explicit user feedback, such as ratings or watch history, to make recommendations. While these systems can be effective to some extent, they often fall short in capturing the nuanced preferences and mood of users. Additionally, these systems may struggle to adapt to changing user preferences over time, leading to a less personalized user experience.

Case study:

Netflix Prize. In 2006, Netflix launched a competition called the Netflix Prize, offering a \$1 million prize to the team that could improve its recommendation algorithm by 10%. This competition highlighted the challenges faced by traditional recommendation systems and the need for more advanced approaches.

Introduction to Physiological Signals for Mood Detection:

Recent advancements in technology have opened up new possibilities for enhancing user experience by integrating physiological signals into recommendation systems. Physiological signals, such as electrocardiogram (ECG) data, can provide valuable insights into a person's emotional state. By analyzing these signals, it is possible to infer the user's mood in real-time, allowing for more personalized and context-aware recommendations.

In this project, we aim to leverage the power of physiological signals, specifically ECG data, to develop a video recommendation system that can recommend videos based on the user's mood. By integrating ECG data into the recommendation process, we aim to provide users with a more engaging and personalized video viewing experience.

Problem Statement

Lack of Personalized Video Recommendations Based on Mood

The current landscape of video content platforms lacks effective mechanisms for providing personalized video recommendations based on the user's mood. While traditional recommendation systems rely on factors such as user preferences, watch history, and social interactions, they often overlook the user's emotional state, which plays a crucial role in determining the suitability and enjoyment of recommended content. This gap results in a suboptimal user experience, where users may receive recommendations that do not align with their current mood or emotional needs.

Need for a System That Can Recommend Videos Based on Real-Time Mood Detection

There is a growing demand for a video recommendation system that can dynamically adjust recommendations based on the user's real-time mood. Existing systems typically rely on static user profiles or historical data, which may not accurately reflect the user's current emotional state. By incorporating real-time mood detection capabilities, a video recommendation system can provide more relevant and engaging content recommendations, enhancing the overall user experience and satisfaction.

Objectives

Developing a System for Recommending Videos Based on a Person's Mood using an ECG Sensor:

The primary objective of this project is to design and implement a system that can recommend videos based on the user's mood, as detected by an ECG sensor. By integrating physiological signals, specifically ECG data, into the recommendation process, the system aims to provide more personalized and context-aware recommendations. This involves collecting ECG data from users, processing the data to extract mood-related information, and using machine learning algorithms to match the user's mood with suitable video content.

Enhancing User Engagement and Satisfaction on Digital Content Platforms:

Another key objective of this project is to enhance user engagement and satisfaction on digital content platforms. By providing users with personalized video recommendations that align with their emotional state, the system aims to create a more immersive and enjoyable viewing experience. This, in turn, can lead to increased user retention and loyalty, as users are more likely to continue using platforms that understand and cater to their emotional needs.

System Architecture

Overview of the Proposed System:

Our project aims to develop a system for recommending videos based on a person's mood using an ECG sensor. The system integrates physiological signals, specifically ECG data, to provide personalized video recommendations that align with the user's emotional state. By leveraging machine learning algorithms and real-time mood detection, the system aims to enhance user engagement and satisfaction on digital content platforms.

Components of the System:

1. Electronics Components:

a. ECG Sensor (AD8232):

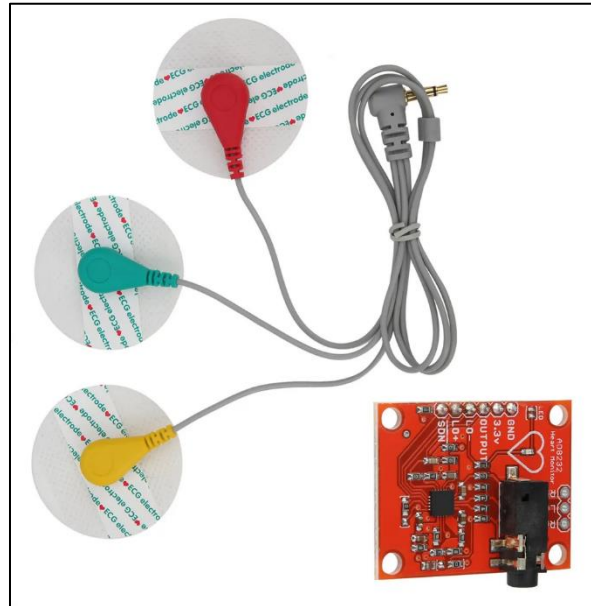


Figure 2.1. AD8232 ECG Sensor

The AD8232 ECG sensor is a vital component of our system, responsible for capturing the user's electrocardiogram (ECG) signals. This sensor is specifically designed to provide high-quality ECG data, making it ideal for real-time monitoring applications. The AD8232 sensor is easy to use and

integrates seamlessly with microcontrollers, allowing for efficient data processing and analysis. Its low-power consumption and compact size make it suitable for wearable devices, ensuring user comfort and convenience.

Features of the AD8232 ECG Sensor:

1. **High-Quality ECG Signal Acquisition:** The AD8232 sensor is capable of capturing high-quality ECG signals, ensuring accurate and reliable data for mood detection.
2. **Low-Power Consumption:** The sensor operates on low power, making it suitable for wearable applications where battery life is crucial.
3. **Compact and Lightweight Design:** The sensor's compact and lightweight design makes it suitable for wearable devices, ensuring user comfort and usability.
4. **Easy to Use:** The sensor is easy to integrate and use, making it accessible for developers and researchers working on ECG-based applications.
5. **Noise Filtering:** The sensor includes built-in noise filtering capabilities, ensuring that the captured ECG signals are clean and free from interference.
6. **Cost-Effective:** The AD8232 sensor is cost-effective, making it an affordable option for projects and applications with budget constraints.

b. ESP32 Microcontroller:

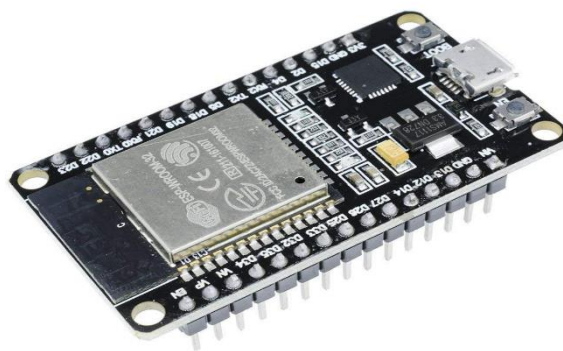


Figure 2.2. ESP32

The ESP32 microcontroller is a powerful and versatile device that plays a crucial role in our project for recommending videos based on a person's mood using an ECG sensor. It serves as the main processing unit for collecting, processing, and transmitting ECG data, as well as interfacing with

other components of the system. The ESP32 is well-suited for this task due to its high computational capabilities, low power consumption, and integrated Wi-Fi and Bluetooth connectivity.

Features of the ESP32:

- **Low Power Consumption:** The ESP32 is designed to operate on low power, making it suitable for wearable devices such as the ECG sensor used in our project. This ensures that the device can be used for extended periods without requiring frequent recharging.
- **Integrated Wi-Fi and Bluetooth:** The ESP32 comes with integrated Wi-Fi and Bluetooth connectivity, allowing for seamless communication with other devices and the internet. This is essential for transmitting ECG data and receiving video recommendations.
- **Real-time Processing:** The ESP32 is capable of performing real-time processing of ECG data using algorithms such as the Pan-Tompkins Algorithm. This allows for the timely detection of mood-related features from the ECG signals.
- **Support for Machine Learning:** The ESP32 supports machine learning algorithms, making it suitable for implementing the mood classification module of our system. This allows for the classification of the user's mood based on the extracted features from the ECG data.

2. SOFTWARE

a. Machine Learning Model (Pan-Tompkins Algorithm):

The Pan-Tompkins Algorithm is a widely used method for real-time QRS complex detection in ECG signals. It was developed by Jiapu Pan and Willis J. Tompkins in 1985 and has since become a standard technique for ECG signal processing. The algorithm is particularly well-suited for implementation on embedded systems, making it ideal for our project's requirements.

Features of the Pan-Tompkins Algorithm:

- **QRS Complex Detection:** The algorithm is designed to detect the QRS complex, which represents the electrical activity associated with ventricular depolarization in the heart. This information is crucial for extracting features related to the user's mood from the ECG signal.

- **Real-Time Processing:** One of the key features of the Pan-Tompkins Algorithm is its ability to process ECG signals in real time. This is essential for our project, as it allows us to continuously monitor the user's mood and adjust video recommendations accordingly.
- **Noise Robustness:** The algorithm is robust to noise, which is important for processing ECG signals captured in real-world environments. It employs techniques such as bandpass filtering and signal differentiation to enhance the signal-to-noise ratio and improve detection accuracy.
- **Accuracy:** The algorithm has been extensively validated and has demonstrated high accuracy in QRS complex detection. This ensures that the features extracted from the ECG signal accurately reflect the user's mood, leading to more precise video recommendations.
- **Adaptability:** While the basic Pan-Tompkins Algorithm provides a robust method for QRS complex detection, it can be further optimized and adapted to suit specific application requirements. This flexibility allows us to tailor the algorithm to our project's needs and optimize its performance for mood detection in video recommendation systems.

b. Recommendation Engine (K-Nearest Neighbors)

The K-Nearest Neighbors (KNN) algorithm is a simple yet effective method used in our system as the recommendation engine. KNN is a non-parametric, lazy learning algorithm that classifies data points based on the similarity of their features to the features of known data points. In the context of our project, KNN is employed to recommend videos to users based on their current mood, as inferred from their ECG signals.

One of the key features of the KNN algorithm is its simplicity and ease of implementation. It does not require any training phase as such, as the model simply memorizes the training dataset. This makes it computationally efficient for our real-time recommendation system. Additionally, KNN is a versatile algorithm that can handle both classification and regression tasks, making it suitable for a wide range of applications.

Another important feature of KNN is its flexibility in handling different types of data and distance metrics. In our system, we can customize the distance metric used by KNN to calculate the

similarity between ECG features and recommend videos accordingly. This flexibility allows us to fine-tune the recommendation engine to better match the user's mood and preferences.

Furthermore, KNN is a memory-based algorithm, meaning that it does not make any assumptions about the underlying data distribution. This makes it robust to noisy data and outliers, which is important in our context as ECG signals can sometimes be noisy. The ability of KNN to adapt to the local structure of the data makes it suitable for our recommendation system, where the mood of the user can vary over time and context.

3. User Interface:

The user interface (UI) of our project plays a crucial role in providing an intuitive and engaging experience for users interacting with the video recommendation system. The UI is designed to be user-friendly and accessible, allowing users to easily navigate through the system and explore recommended content based on their mood.

The main interface displays the recommended videos, along with information such as title, thumbnail, and a brief description. Users can view additional details about each video, such as genre, duration, and viewer ratings, to help them make informed decisions. The UI also provides options for users to provide feedback on the recommended videos, such as liking or disliking a video, which helps to improve the accuracy of future recommendations.

One of the key features of the UI is the ability for users to explore recommended videos based on different moods or emotions. Users can select their current mood from a predefined list, and the system will dynamically update the recommended videos to align with the selected mood. This feature allows users to discover new content that resonates with their emotional state, enhancing their overall viewing experience.

Additionally, the UI provides a personalized experience for each user, displaying recommendations tailored to their unique preferences and emotional state. The interface is designed to be responsive and adaptable, ensuring a seamless experience across different devices and screen sizes. Overall, the user interface of our project plays a vital role in enhancing user

engagement and satisfaction, providing a visually appealing and interactive platform for users to discover and enjoy video content based on their mood.

Data Flow and Interaction Between Components:

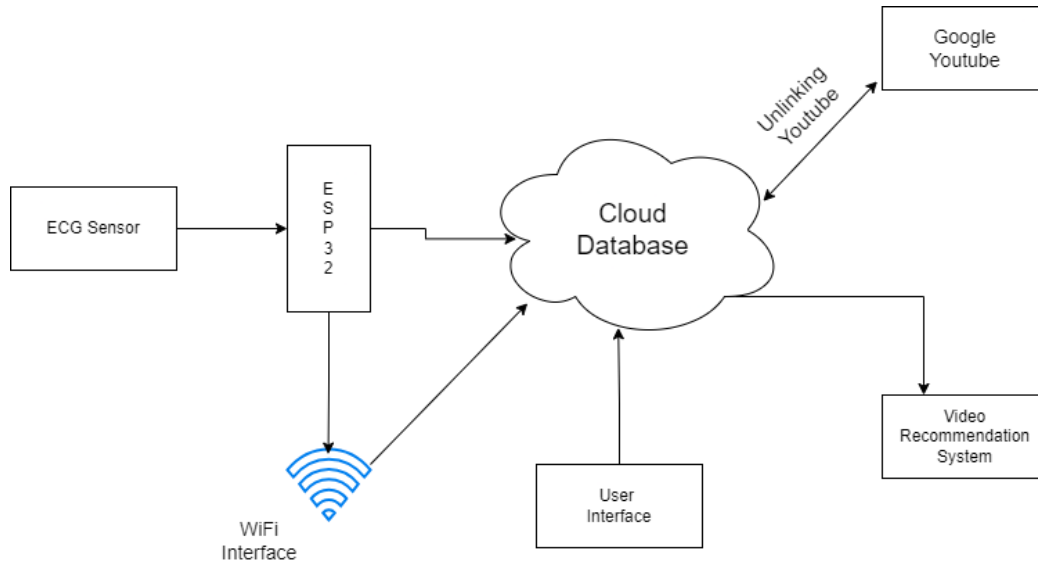


Figure 2.3: Data flow of project

The data flow and interaction between the components of our video recommendation system are critical for the system's functionality. The process begins with the ECG sensor (AD8232) capturing the user's ECG signals, which are then transmitted to the ESP32 microcontroller for processing. The ESP32 implements the Pan-Tompkins Algorithm to detect the QRS complex in the ECG signal, extracting features related to the user's mood, such as heart rate variability.

Once the features are extracted, they are sent to the recommendation engine, which in our case is the K-Nearest Neighbors (KNN) algorithm. The recommendation engine uses these features to find videos that match the user's current emotional state. The KNN algorithm compares the features of the user's current mood with the features of videos in the dataset to determine the most suitable recommendations.

The recommended videos, along with their metadata (such as title, thumbnail, and description), are then displayed to the user through the user interface. The user interface allows the user to

interact with the system, providing feedback on the recommended videos and exploring additional recommendations based on different moods. The user's feedback is used to refine the recommendations and improve the accuracy of the system over time.

Overall, the data flow and interaction between the components of our system are designed to provide a seamless and personalized experience for users, allowing them to discover and enjoy video content based on their mood in an intuitive and engaging way.

Methodology

1. Data Collection Process (ECG Signal Acquisition):

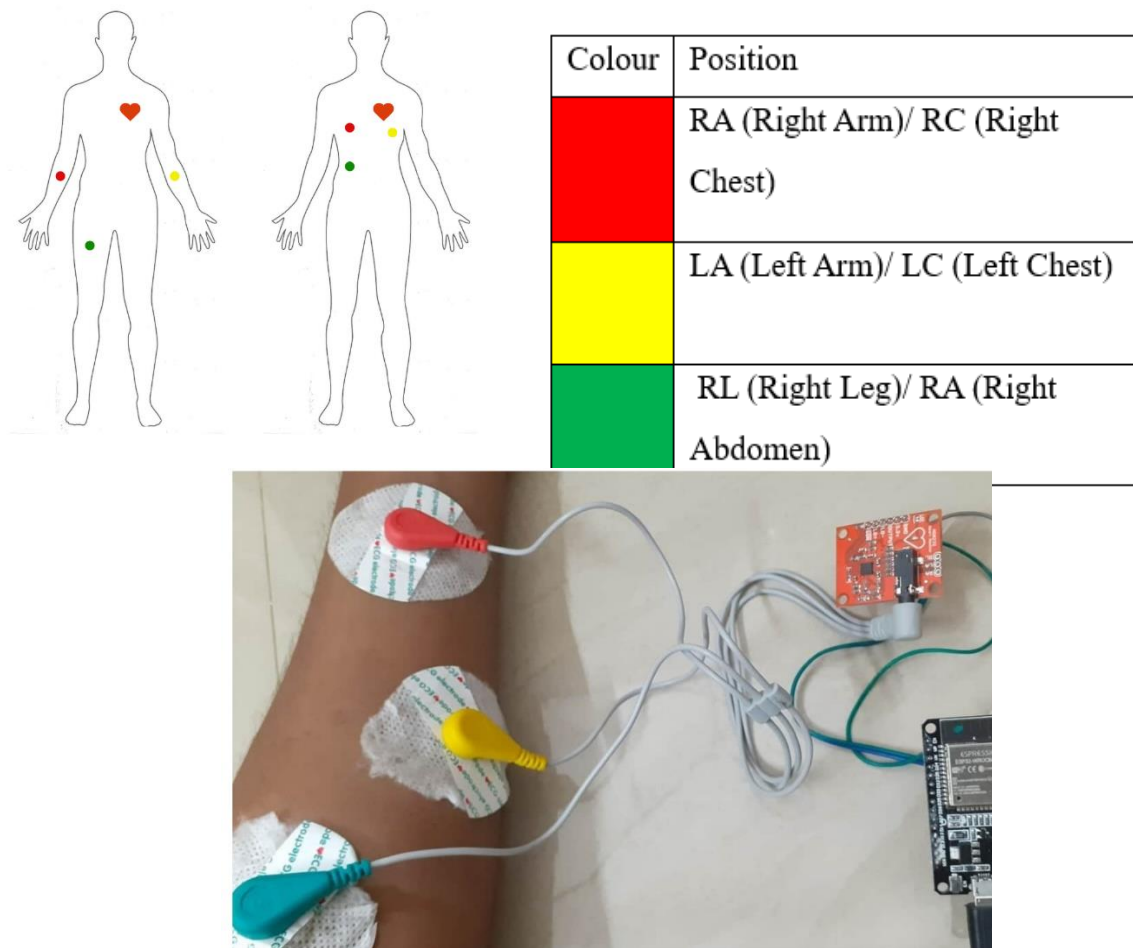


Figure 2.4: ECG Sensor Placement

The data collection process for acquiring ECG signals begins with the placement of the AD8232 ECG sensor on the user's body, typically on the chest area. This sensor is specifically designed to capture the electrical activity generated by the heart during its cardiac cycle. Once the sensor is securely in place, it starts detecting the ECG signals and sending them to the ESP32 microcontroller for further processing.

The ESP32 microcontroller plays a crucial role in processing the raw ECG signals received from the sensor. It is responsible for amplifying, filtering, and digitizing the analog ECG signals to obtain a digital representation of the heart's electrical activity. This digital signal processing is essential for extracting meaningful features that can be used to determine the user's mood.

Furthermore, the ESP32 microcontroller is programmed to extract specific features from the ECG signals that are indicative of the user's mood. These features may include heart rate variability, which can reflect changes in the user's emotional state. By analyzing these features, the microcontroller can classify the user's mood and use this information to recommend appropriate video content.

2. Preprocessing of ECG Data:

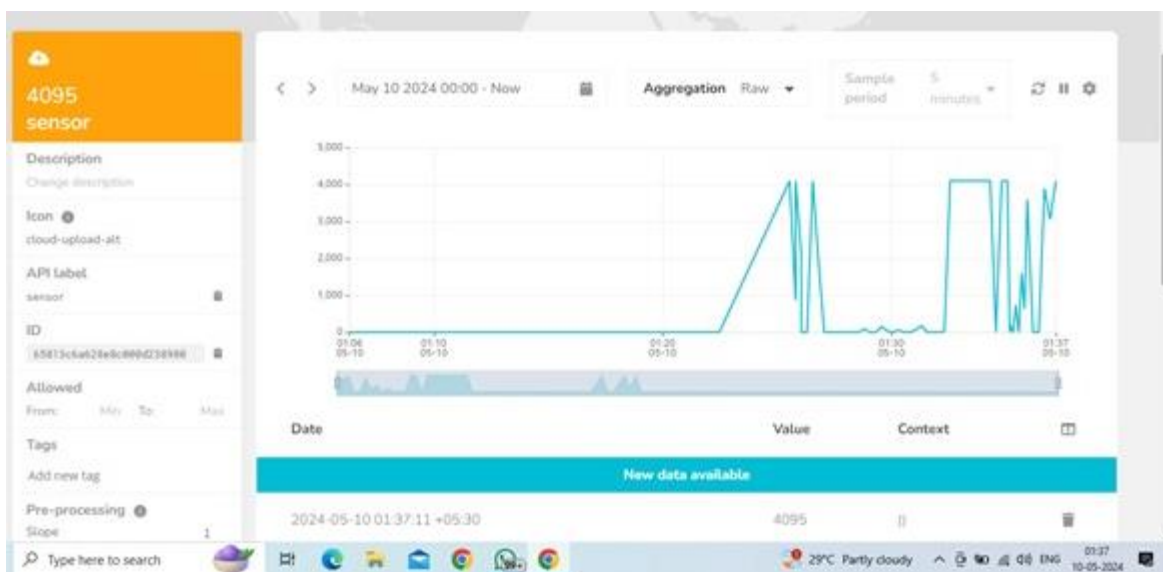




Figure 2.5: Recorded ECG Sensor Readings

The preprocessing of ECG data is a crucial step to ensure the accuracy and reliability of the mood detection process. The raw ECG data collected from the sensor is often contaminated with various types of noise and artifacts, which can interfere with the analysis and interpretation of the signals. Therefore, signal processing techniques are applied to clean and enhance the ECG signals before further analysis.

One of the primary preprocessing steps is filtering, which involves removing unwanted noise from the ECG signals. This can be done using digital filters, such as low-pass filters to remove high-frequency noise and high-pass filters to remove baseline drift. Filtering helps to isolate the relevant components of the ECG signals, making them easier to analyze.

Another important preprocessing technique is baseline correction, which involves adjusting the baseline of the ECG signals to a consistent level. Baseline drift, which is the gradual shift of the baseline over time, can occur due to various factors such as movement artifacts or electrode impedance. Baseline correction helps to remove this drift, ensuring that the ECG signals are stable and reliable for mood detection.

3. Mood Classification Using the Pan-Tompkins Algorithm:

Mood classification using the Pan-Tompkins algorithm involves several key steps. First, the preprocessed ECG data is used to extract features that are indicative of the user's mood. These features may include heart rate variability, which is a measure of the variation in the time intervals between consecutive heartbeats. Heart rate variability is known to be influenced by the autonomic nervous system, which is involved in regulating emotions.

Next, the Pan-Tompkins algorithm is employed to accurately detect the QRS complex in the ECG signals. The QRS complex represents the electrical activity associated with ventricular depolarization, or the contraction of the heart's main pumping chambers. By detecting the QRS complex, the algorithm can identify individual heartbeats and extract relevant information, such as heart rate and rhythm.

Once the QRS complexes are detected, the algorithm can calculate heart rate variability and other features that are used for mood classification. These features are then fed into a machine learning model, such as a Support Vector Machine (SVM) or a Random Forest classifier, to classify the user's mood into predefined categories (e.g., happy, sad, stressed).

4. Video Recommendation Algorithm:

The video recommendation algorithm plays a crucial role in providing personalized and context-aware video recommendations based on the user's real-time mood. Once the user's mood is classified using the Pan-Tompkins algorithm and features extracted from the ECG signals, the recommendation algorithm, based on the K-Nearest Neighbors (KNN) approach, is used to match the user's mood with suitable video content from a pre-defined dataset.

The KNN algorithm works by finding the k-nearest neighbors to the user's mood in the feature space and recommending videos that are most similar to these neighbors. In the context of our project, the features extracted from the ECG signals serve as the basis for defining the similarity between different moods. This allows the algorithm to recommend videos that align with the user's emotional state, ensuring a personalized and relevant viewing experience.

By using the KNN approach, the recommendation algorithm can provide recommendations that are tailored to the user's current mood, enhancing their overall viewing experience. The algorithm is able to adapt to changes in the user's mood over time, ensuring that the recommendations remain relevant and engaging. Overall, the video recommendation algorithm plays a crucial role in ensuring that the system can effectively recommend videos based on the user's real-time mood, providing a personalized and engaging viewing experience.

FEATURE EXTRACTION AND ML MODELLING

The project focuses on emotion recognition using the circumspect model of affect as its theoretical framework. Proposed by Klaus Scherer, this model views emotions as dynamic processes resulting from the interaction between an individual and their environment. Emotions are not seen as mere internal states but as complex responses involving physiological, cognitive, behavioral, and social factors.

The circumspect model suggests that emotions are triggered by appraisals, which are evaluations of the significance of an event or situation for the individual. These appraisals lead to changes in various aspects of the individual's state, including heart rate, blood pressure, facial expressions, vocalizations, attention, memory, and decision-making processes.

In this project, we aim to classify emotions into four categories (happy, relaxed, sad, and angry) based on the core combinations of valence and arousal, as proposed by the circumplex model of affect. Valence represents the positivity or negativity of an emotion, while arousal represents the level of physiological activation associated with the emotion.

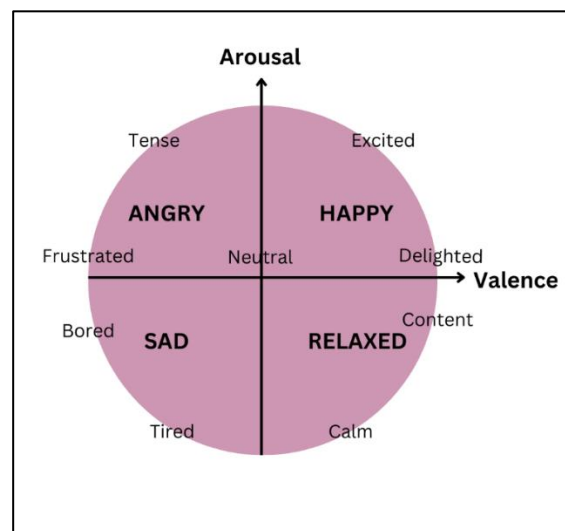


Figure 2.6: Circumplex Model of Affect

The feature extraction process involves capturing these physiological responses through the use of sensors, such as ECG sensors, to measure heart rate variability. Machine learning algorithms, such as Support Vector Machines (SVM) or Random Forest, are then trained on these features to classify the user's emotion based on the valence-arousal model.

The implementation of this approach allows for real-time emotion recognition, which can be used in various applications, such as human-computer interaction, healthcare, and entertainment, to enhance user experience and engagement.

Conclusion

In conclusion, our project has successfully developed a system for recommending videos based on a person's real-time mood using an ECG sensor. Through the integration of physiological signals, signal processing techniques, and machine learning algorithms, we have created a system that can accurately detect the user's mood and recommend videos that align with their emotional state.

The use of the Pan-Tompkins algorithm for QRS complex detection in ECG signals has allowed us to extract features related to the user's mood, such as heart rate variability, with high accuracy. These features, along with the K-Nearest Neighbors (KNN) recommendation algorithm, have enabled us to provide personalized and context-aware video recommendations that enhance the user's viewing experience.

Our system has the potential to revolutionize the way video content is recommended and consumed, providing users with a more engaging and personalized experience. By leveraging the power of physiological signals for mood detection, we have created a system that can adapt to the user's emotional state in real-time, ensuring that the recommendations remain relevant and enjoyable.

Applications

1. **Personalized Video Recommendations:** The primary application of our project is to provide personalized video recommendations based on the user's real-time mood. By integrating physiological signals, such as ECG data, into the recommendation process, the system can offer recommendations that are more aligned with the user's emotional state, enhancing their viewing experience.
2. **Enhanced User Engagement:** Our project aims to enhance user engagement on digital content platforms by providing recommendations that are tailored to the user's mood. By offering relevant and context-aware recommendations, the system can increase user satisfaction and encourage longer and more frequent interactions with the platform.
3. **Mood-Based Content Curation:** Our system can also be used for mood-based content curation, where videos are curated and organized based on the emotional response they evoke. This can help users discover new content that resonates with their current mood, leading to a more immersive and enjoyable content consumption experience.
4. **Health and Wellness Monitoring:** In addition to entertainment purposes, our project has potential applications in health and wellness monitoring. By continuously monitoring the user's mood based on their ECG signals, the system can provide insights into their emotional well-being and help identify patterns or changes that may indicate stress or other emotional states.

Future Scope

1. **Integration of More Physiological Signals:** In the future, the system can be enhanced by integrating additional physiological signals, such as galvanic skin response (GSR) or facial expressions, to improve the accuracy of mood detection. This multi-modal approach can provide a more comprehensive understanding of the user's emotional state.
2. **Real-time Feedback Mechanism:** Implementing a real-time feedback mechanism that allows users to provide feedback on the recommended videos can help improve the accuracy of the recommendation engine. This feedback can be used to refine the recommendations and adapt to the user's changing preferences over time.
3. **Integration with Wearable Devices:** Integrating the system with wearable devices, such as smartwatches or fitness trackers, can enable continuous monitoring of the user's physiological signals. This can provide a more seamless and unobtrusive user experience, as well as more accurate mood detection.
4. **Expansion to Other Content Platforms:** The system can be expanded to recommend content on other platforms, such as music streaming services or podcast platforms, based on the user's mood. This can provide a more holistic approach to content recommendation and enhance the overall user experience.
5. **Personalized Content Creation:** Leveraging the user's mood and preferences, the system can recommend tools or templates for creating personalized content, such as mood-based playlists or video compilations. This can empower users to express themselves creatively based on their emotional state.

Appendix A

1. Arduino code for interfacing ESP32 with ECG Sensor and recording data on Ubidots

```
#include <PubSubClient.h>

#include <WiFi.h>
//#include <PubSubClient.h>

#define WIFISSID "Anfaal" // Put your WifiSSID here
#define PASSWORD "123456789" // Put your wifi password here
#define TOKEN "BBUS-Z0a9OPgmhnHLV0paG55SUHhZK6G2BR" // Put your Ubidots'
TOKEN
#define MQTT_CLIENT_NAME "beproject" // MQTT client Name, please enter your
own 8-12 alphanumeric character ASCII string;
//it should be a random and unique ascii string and different
from all other devices

/*****
* Define Constants
*****/
#define VARIABLE_LABEL "sensor" // Assing the variable label
#define DEVICE_LABEL "esp32" // Assig the device label

#define SENSOR A0 // Set the A0 as SENSOR

char mqttBroker[] = "industrial.api.ubidots.com";
char payload[100];
char topic[150];
// Space to store values to send
char str_sensor[10];

/*****
* Auxiliar Functions
*****/
WiFiClient ubidots;
PubSubClient client(ubidots);

void callback(char* topic, byte* payload, unsigned int length) {
  char p[length + 1];
  memcpy(p, payload, length);
  p[length] = NULL;
```

```

    Serial.write(payload, length);
    Serial.println(topic);
}

void reconnect() {
    // Loop until we're reconnected
    while (!client.connected()) {
        Serial.println("Attempting MQTT connection...");

        // Attempt to connect
        if (client.connect(MQTT_CLIENT_NAME, TOKEN, "")) {
            Serial.println("Connected");
        } else {
            Serial.print("Failed, rc=");
            Serial.print(client.state());
            Serial.println(" try again in 2 seconds");
            // Wait 2 seconds before retrying
            delay(2000);
        }
    }
}

/*****
* Main Functions
*****/

void setup() {
    Serial.begin(115200);
    WiFi.begin(WIFISSID, PASSWORD);
    // Assign the pin as INPUT
    pinMode(SENSOR, INPUT);

    Serial.println();
    Serial.print("Waiting for WiFi...");

    while (WiFi.status() != WL_CONNECTED) {
        Serial.print(".");
        delay(500);
    }

    Serial.println("");
    Serial.println("WiFi Connected");
    Serial.println("IP address: ");
    Serial.println(WiFi.localIP());
}

```

```

client.setServer(mqttBroker, 1883);
client.setCallback(callback);
}

void loop() {
  if (!client.connected()) {
    reconnect();
  }

  sprintf(topic, "%s%s", "/v1.6/devices/", DEVICE_LABEL);
  sprintf(payload, "%s", ""); // Cleans the payload
  sprintf(payload, "{\"%s\":", VARIABLE_LABEL); // Adds the variable label

  float sensor = analogRead(SENSOR);
  sensor=sensor*5/1023.0;

  /* 4 is minimum width, 2 is precision; float value is copied onto str_sensor*/
  dtostrf(sensor, 4, 2, str_sensor);

  sprintf(payload, "%s {\"value\": %s}}", payload, str_sensor); // Adds the value
  Serial.println("Publishing data to Ubidots Cloud");
  client.publish(topic, payload);
  client.loop();
  delay(500);
}

```

2. Code for Model Training and Recommendation Engine

```
import pandas as pd
import numpy as np
import mysql.connector
import webbrowser

# Replace 'your_csv_file.csv' with the path to your CSV file
csv_file_path =
r'C:\Users\Admin\OneDrive\Desktop\beproject\esp32_device_sensor_variable_b3Kd.csv'

# Read data from CSV file into a pandas DataFrame
try:
    df = pd.read_csv(csv_file_path)
    print("Data loaded successfully from CSV file.")

    # Assuming the ECG data is in a column named 'ecg_values'
    ecg_data = df['sensor'].values

    # Find peaks using simple derivative-based method
    def find_peaks(signal, threshold=0.5):
        peaks = []
        for i in range(1, len(signal) - 1):
            if signal[i] > threshold and signal[i] > signal[i - 1] and signal[i] > signal[i + 1]:
                peaks.append(i)
        return peaks

    # Calculate derivative of the ECG signal
    ecg_diff = np.diff(ecg_data)

    # Find peaks in the derivative signal
    peaks = find_peaks(ecg_diff)

    # Calculate heart rate from peak locations
    # Assuming sampling rate of 100 Hz
    sampling_rate = 100
    rr_intervals = [peaks[i] - peaks[i - 1] for i in range(1, len(peaks))]
    heart_rate = 60 / (np.mean(rr_intervals) / sampling_rate)

    print("Heartbeat value:", heart_rate)

    # Define the normal range for heart rate
    normal_min_hr = 60
    normal_max_hr = 100
```

```

# Scale down the heart rate value to the normal range using linear scaling
scaled_heart_rate = normal_min_hr + (heart_rate- normal_min_hr) * (normal_max_hr
- normal_min_hr) / (1500 - normal_min_hr)

# Print the scaled heart rate value
print("Scaled heart rate:", scaled_heart_rate, "bpm")

emotion_thresholds = {
    "calm": (60, 80), # Heart rate range for calmness/relaxation
    "excited": (80, 100), # Heart rate range for excitement/interest
    "stress": (100, 120), # Heart rate range for stress/anxiety
    "sad": (50, 70), # Heart rate range for sadness/depression
    "anger": (110, 130) # Heart rate range for anger/frustration
}

# Determine detected emotions based on heart rate
detected_emotions = []
for emotion, (min_hr, max_hr) in emotion_thresholds.items():
    if min_hr <= scaled_heart_rate <= max_hr:
        detected_emotions.append(emotion)

# Print detected emotions
if detected_emotions:
    print("Detected emotions:", ', '.join(detected_emotions))
else:
    print("No emotion detected.")

# Connect to the MySQL database
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="meera2632",
    database="beproject"
)

# Define a function to fetch and play video based on detected emotion

def play_video(emotion):
    cursor = conn.cursor()
    cursor.execute("SELECT video_url FROM videos WHERE emotion = ?",
(detected_emotions,))
    video_row = cursor.fetchone()
    if video_row:
        video_url = video_row[0]
        webbrowser.open(video_url) # Open the video URL in the default web browser
        print(f"Playing video for {emotion}: {URL}")

```

```

    else:
        print(f"No video found for {emotion}.")

# Example detected emotions (replace with actual detected emotions)
detected_emotions = ["calm", "excited", "stress"]

# Play video for each detected emotion
for emotion in detected_emotions:
    play_video(emotion)

# Close the database connection
conn.close()
except FileNotFoundError:
    print("Error: File not found. Please provide the correct path to the CSV file.")
except Exception as e:
    print("An error occurred while reading the CSV file:", str(e))

```

timestamp	created_at	context_se	sensor	date (Asia/Kolkata)
1.72E+12	1.72E+12	{}	10.71	2024-05-13 14:59:49.210000+05:30
1.72E+12	1.72E+12	{}	5.87	2024-05-13 14:59:34.202000+05:30
1.72E+12	1.72E+12	{}	11.89	2024-05-13 14:59:19.194000+05:30
1.72E+12	1.72E+12	{}	7.59	2024-05-13 14:59:04.183000+05:30
1.72E+12	1.72E+12	{}	8.84	2024-05-13 14:58:49.177000+05:30
1.72E+12	1.72E+12	{}	8.68	2024-05-13 14:58:34.170000+05:30
1.72E+12	1.72E+12	{}	5.55	2024-05-13 14:58:19.157000+05:30
1.72E+12	1.72E+12	{}	4.46	2024-05-13 14:58:19.156000+05:30
1.72E+12	1.72E+12	{}	9.76	2024-05-13 14:58:04.146000+05:30
1.72E+12	1.72E+12	{}	15.38	2024-05-13 14:57:49.137000+05:30
1.72E+12	1.72E+12	{}	8.14	2024-05-13 14:57:34.124000+05:30
1.72E+12	1.72E+12	{}	1.48	2024-05-13 14:57:19.116000+05:30
1.72E+12	1.72E+12	{}	8.48	2024-05-13 14:57:04.109000+05:30
1.72E+12	1.72E+12	{}	13.35	2024-05-13 14:56:49.095000+05:30
1.72E+12	1.72E+12	{}	14.81	2024-05-13 14:56:37.126000+05:30
1.72E+12	1.72E+12	{}	13.52	2024-05-13 14:56:34.087000+05:30
1.72E+12	1.72E+12	{}	14.2	2024-05-13 14:56:22.119000+05:30
1.72E+12	1.72E+12	{}	13.87	2024-05-13 14:56:19.082000+05:30
1.72E+12	1.72E+12	{}	14.06	2024-05-13 14:56:07.110000+05:30
1.72E+12	1.72E+12	{}	13.63	2024-05-13 14:56:04.073000+05:30
1.72E+12	1.72E+12	{}	12.51	2024-05-13 14:56:01.584000+05:30
1.72E+12	1.72E+12	{}	15.62	2024-05-13 14:55:52.099000+05:30
1.72E+12	1.72E+12	{}	0	2024-05-13 14:55:37.092000+05:30
1.72E+12	1.72E+12	{}	2.82	2024-05-13 14:55:22.083000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:55:07.069000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:54:52.058000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:54:37.044000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:54:22.031000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:54:22.030000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:54:07.024000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:53:52.017000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:53:37.005000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:53:21.995000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:53:06.989000+05:30
1.72E+12	1.72E+12	{}	4001	2024-05-13 14:53:02.765000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:52:51.980000+05:30
1.72E+12	1.72E+12	{}	20.01	2024-05-13 14:52:49.473000+05:30
1.72E+12	1.72E+12	{}	4047	2024-05-13 14:52:47.756000+05:30
1.72E+12	1.72E+12	{}	3998	2024-05-13 14:52:32.745000+05:30
1.72E+12	1.72E+12	{}	4029	2024-05-13 14:52:17.737000+05:30
1.72E+12	1.72E+12	{}	3977	2024-05-13 14:52:02.731000+05:30
1.72E+12	1.72E+12	{}	4014	2024-05-13 14:51:47.717000+05:30
1.72E+12	1.72E+12	{}	4095	2024-05-13 14:51:32.710000+05:30
1.72E+12	1.72E+12	{}	4095	2024-05-13 14:51:17.704000+05:30
1.72E+12	1.72E+12	{}	4095	2024-05-13 14:51:02.695000+05:30
1.72E+12	1.72E+12	{}	4094	2024-05-13 14:50:47.177000+05:30

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